Structural Inductive Biases for Neural Conservation Laws: Contributions

Follow-up to Chen, Gelb & Lee (2023) — Conservative Form Networks. This paper extends the original CFN methodology in five distinct directions, each addressing a limitation or open question left by the original work.

1. Data-Efficient Training: Single Trajectory + Random Windowing

Original work: Training data consists of N_{traj} = 200 independent trajectories, each from a perturbed initial condition. Requires ~200 expensive simulations per experiment.

This work: Training uses a single (or few) trajectory, with random space-time windows extracted during each epoch. The same underlying trajectory provides diverse local state configurations through windowing, eliminating the need for many independent simulations.

Significance: Reduces simulation cost by ~100× while preserving generalization. Makes CFN training practical for problems where each simulation is expensive (geophysical flows, high-resolution fluid dynamics, costly forward models).

2. Modern Baselines: Rollout-Trained Neural Operators

Original work: Baselines are MLPs — nCFN (no conservation) and nCFN-reg (conservation as regularizer). Pre-2020 architecture, 5 hidden layers, 64 nodes.

This work: Baselines include rollout-trained Conv1D networks (CNN-strong, kernel=5, ~85K parameters), rollout-trained Fourier Neural Operator (FNO-strong, 64 modes, ~591K parameters), and conservation-regularized FNO (FNO + mass-conservation penalty in loss).

Significance: Beating modern neural operators is a substantially stronger empirical claim than beating MLPs. The conservation-regularized FNO baseline directly tests whether architectural conservation (CFN's approach) is necessary or whether loss-based regularization suffices — a question the original work does not address.

3. New Generalization Tests: Spatial Extrapolation + Long-Time Shock Propagation

Original work: Tests time extrapolation of 3–10× training horizon on fixed spatial domain. Same N_x for training and testing.

This work: Three distinct generalization tests, none in the original paper:

- **Spatial extrapolation** on Whitham wave-mean: train on N_x=500 (domain length 200 units), evaluate on N_x=20,001 (40× longer domain, same dx). CFN achieves 1.7×10^{-4}; FNO degrades to 6.6×10^{-2}.
- **Long-time prediction through shock formation** on Burgers: train pre-shock, predict post-shock. Tests whether the conservation form gives correct Rankine-Hugoniot shock speed in extrapolation.
- **Long-time prediction with shock + rarefaction** on Saint-Venant: train on short window, predict full wave evolution over 20× training time. Tests vector-system generalization.

Significance: Demonstrates the structural prior gives bounded error along three distinct generalization axes (spatial, temporal-with-shock, multi-field). Far broader scope than the original short-horizon time-extrapolation tests.

4. Theoretical Error Decomposition

Original work: Empirical demonstration only. No analytical explanation of when or why CFN generalizes beyond the training distribution.

This work: A per-step error bound of the form $||u^N_{theta} - u^N_{true}||_{inf} \leq N \cdot (dt/dx) \cdot epsilon \cdot max|du| + O(dt^2)$, where $epsilon = ||F'_{theta} - F'_{true}||_{inf}$ is the static flux derivative error. The bound is **domain-length independent**, explaining the spatial extrapolation result. Empirically verified to predict observed RMSE within a small constant factor across all PDEs.

Significance: Converts CFN from a method that ‘empirically works’ to one with predictive analytical guarantees. Shows no analogous bound exists for FNO (no separable static-flux + integrator decomposition), explaining the structural difference rather than just measuring it.

5. New Physical Context: Modulation Theory and Dispersive Hydrodynamics

Original work: Tested on Burgers, Saint-Venant (shallow water), and Euler (gas dynamics) — classical compressible flow and shock problems.

This work: Adds the Whitham wave-mean equation — a modulation-theoretic conservation law arising in dispersive hydrodynamics (water waves, BEC, nonlinear optics). Demonstrates CFN handles fluxes derived from elliptic-integral averages, beyond the polynomial-flux setting of the original paper.

Significance: Establishes CFN's applicability to a new physical domain. Modulation systems describe slowly-varying envelopes of nonlinear waves — learning them is a step toward predicting dispersive shock wave (DSW) dynamics from limited data, with applications in fluid experiments where full DNS is prohibitive.

Contribution	Original CFN	This Work
Training data	N=200 trajectories	Single trajectory + windowing (~100x less)
Baselines	MLPs (nCFN, nCFN-reg)	FNO, CNN, FNO+conservation-reg
Time horizon	3-10x training time	20-40x training time
Spatial test	Same Nx as training	40x longer domain (zero-shot)
Theory	Empirical only	Per-step error bound, domain-length independent
PDE scope	Burgers, SV, Euler	Wave-mean (new), Burgers, Saint-Venant